



Effects of the periparturient period and parity on local innervation markers in subcutaneous adipose tissue of dairy cows

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ABSTRACT

Adipose tissue serves as a vital energy source to meet the metabolic demands of the periparturient period in dairy cows. However, excessive release of nonesterified fatty acids (NEFA) from adipose tissue can lead to postpartum metabolic diseases. The objectives of this investigation were to identify changes in expression of nervous system gene and protein markers in the adipose tissue of dairy cows at varying parities during the periparturient period and to define their associations with circulating markers of metabolism and inflammation. Tailhead subcutaneous adipose tissue (SAT) was collected from 16 multiparous Holstein dairy cows (3.5 ± 1.6 lactations, $15,136 \pm 1,743$ previous 305-d milk yield, mean $\pm$ SD) approximately 10 d before parturition (8.56 ± 2.3 d to parturition, mean $\pm$ SD) and at 10 d postpartum. Blood samples were collected concurrently with SAT biopsy at prepartum time point and again, separately, at 7 d postpartum. Blood metabolites and inflammatory markers were quantified, and SAT was analyzed for markers related to nervous system signaling. Differences were assessed using a repeated-measures mixed-effects model in PROC MIXED in SAS (version 9.4; SAS Institute Inc., Cary, NC). Period was treated as a repetition by cow nested in parity, and parity, period, and the interaction of parity and period served as fixed effects. Metabolic profiling was consistent with that of transition dairy cows, with elevated NEFA and BHB and decreased insulin concentrations in the postpartum period. Approximately 56% of the cows demonstrated elevated lipolysis (EL) in the postpartum period, and 44% of animals developed disease in the first 7 DIM. The expression of catecholamine biosynthesis markers tyrosine hydroxylase (TH), phosphorylated TH serine 40 residue (pTH_{ser40}), dopamine β -hydroxylase (DBH), and norepinephrine transporter (NET) did not change in SAT

from the prepartum to postpartum period, suggesting a stable catecholamine biosynthesis machinery during the periparturient period. However, protein abundance of TH, pTH_{ser40}, protein gene product 9.5, NET, nerve growth factor receptor (intracellular domain 25 kDa), and glial fibrillary acidic protein increased with parity. Interestingly, DBH, the enzyme that converts dopamine to norepinephrine, decreased abundance with parity, suggesting reduced local capacity for norepinephrine synthesis in the SAT of older cows. Concurrently, increased expression of NET may reflect altered sympathetic signaling dynamics in SAT with advancing parity. The sensory system marker, calcitonin gene-related peptide, was detectable in SAT but did not differ across the periparturient period and was more abundant in cows in the third lactation. These findings indicate that advancing parity is associated with expansion and reorganization of local SAT innervation rather than acute periparturient remodeling. Markers of lipogenesis and lipolysis regulation but not of adipose tissue being were altered between the prepartum and postpartum periods, suggesting that neural remodeling may shape local SAT metabolic adaptation rather than adipocyte fate. Overall, our results reveal a previously underappreciated role of parity in shaping neural-catecholamine regulation of SAT during the periparturient period, providing new insights into mechanisms that may influence adipose tissue dysfunction and subsequent susceptibility to develop metabolic disorders in multiparous dairy cows.

Key words: periparturient period, nervous system, adipose tissue

INTRODUCTION

Enhanced lipolysis in adipose tissue is one of the main adaptive mechanisms to periparturient negative energy balance in dairy cows (Contreras and Sordillo, 2011). This process is classically regulated by endocrine factors, including glucocorticoids, prolactin, and growth hormone (Vernon et al., 1995); however, initiation and magnitude

Received August 29, 2025.

Accepted February 28, 2026.

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

of lipolysis across mammalian species is primarily regulated by the sympathetic nervous system (SNS) through β -adrenergic receptor signaling (ADRB; McNamara and Hillers, 1986; Bartness et al., 2014; Martinez-Sanchez et al., 2022). Excessive or dysregulated lipid mobilization during early lactation contributes to elevated circulating nonesterified fatty acids (NEFA), inflammation, and increased risk of postpartum metabolic disease, underscoring the importance of dissecting mechanisms that regulate adipose tissue lipolysis in dairy cows.

In rodents, local SNS innervations of adipose tissue plays a critical role not only in lipolysis but also in adipose tissue remodeling, including neural plasticity and, in some depots, induction of beige adipocytes (beiging), capable of nonshivering thermogenesis through catecholamines release (Bartness et al., 2010a,b; Luo and Liu, 2016; Sakers et al., 2022). Although brown and beige adipose tissues dissipate energy through oxidative phosphorylation (Cheng et al., 2021), white adipose tissue serves as the primary lipid storage depot and an active endocrine organ (Luo and Liu, 2016). In dairy cows, adipose tissue beiging remains poorly characterized and is traditionally thought to be minimal after birth (Alexander et al., 1975). Nonetheless, prior evidence in ruminants suggests that adipose tissue remodeling during metabolic stress may involve pathways beyond classical lipolysis (Asano et al., 2013).

As dairy cows transition from late gestation to lactation, adipose tissue undergoes extensive remodeling, including lipolysis, local inflammation, and changes in the extracellular matrix organization (Contreras et al., 2017). These processes are likely influenced by local neuroregulatory mechanisms. In addition to SNS-mediated adrenergic signaling, sensory nerves may provide feedback on tissue metabolic and inflammatory status, and neural plasticity, including axonal growth, regression, and myelination, may shape adipose tissue responsiveness during periods of metabolic challenge. However, in periparturient dairy cows, current knowledge of adipose tissue innervation is limited to a small branch of the SNS, describing the binding affinity, density, and sensitivity of adrenergic receptors (Jaster and Wegner, 1981; McNamara and Hillers, 1986; Sumner and McNamara, 2007), leaving broader aspects of neural remodeling and catecholamine handling unexplored.

Notably, components of the neurotrophic system have been identified in subcutaneous adipose tissue (SAT) during midlactation in dairy cows (Colitti et al., 2015), suggesting a paracrine neural-metabolic crosstalk in the regulation of adipocyte function (Colitti et al., 2015). Yet whether adipose tissue innervation undergoes dynamic remodeling during the periparturient period and whether this remodeling differs with parity remains unknown. Given that parity is a major determinant of metabolic

adaptation and disease risk in dairy cows, understanding how neural regulation of adipose tissue adapts with advancing lactation number is particularly relevant.

The rationale for this study was that remodeling of adipose tissue innervation during the periparturient period may determine whether lipid mobilization remains adaptive or progresses toward excessive lipolysis, inflammation, and metabolic dysfunction. Therefore, the objective of this study was to define changes in nervous system-related gene and protein markers in tailhead SAT of varying parity periparturient dairy cows and to evaluate their associations with systemic indicators of metabolism and inflammation. We hypothesized that markers of adipose tissue innervation and neural plasticity increase from the prepartum (PRE) to the postpartum (POST) period, reflecting local neural remodeling that supports metabolic adaptation to lactation. Elucidating these mechanisms will advance the understanding of adipose tissue neuroregulation in dairy cows and may reveal novel molecular and cellular targets to improve metabolic health, welfare, and productivity.

MATERIALS AND METHODS

Animals and Experimental Design

This study was developed using a subset of cows enrolled in a previously published study (Lima et al., 2025). The Institutional Animal Care and Use Committee at Virginia Tech approved all procedures for this study (protocol no. 22-071). Eighty-seven multiparous Holstein dairy cows were enrolled approximately -21 d relative to parturition (-21 ± 3.3 d, mean $\pm$ SD) at the Virginia Tech Dairy Science Complex and remained on the trial until 30 DIM (Lima et al., 2025). Due to adipose tissue sample availability, 16 multiparous cows (3.5 ± 1.6 lactations, $15,136 \pm 1,743$ previous 305-d milk yield, mean $\pm$ SD) with both prepartum and postpartum adipose tissue samples were used for this exploratory study. Blood samples and subcutaneous tailhead adipose tissue samples were taken from each animal before and after calving.

All cows received the same close-up diet from -21 DIM until calving (1.59 Mcal/kg of DM and 14.6% CP) and a lactation diet from calving until 30 DIM (1.82 Mcal/kg of DM and 18.4% CP). Cows were fed once a day (~0800 h) during the close-up and early lactation periods and milked twice a day at approximately 0100 and 1200 h. More detailed information regarding diet composition has been recently reported (Lima et al., 2025).

Body condition score (scale of 1 = thin to 5 = obese, with quarter-point increments) was recorded weekly by 2 trained individuals, and the average score was used for statistical analysis. Rectal temperature (GLA M900

Thermometer, GLA Agricultural Electronics, CA) and blood ketone concentration (Precision Xtra β -ketone, Abbott Diabetes Care) measurements were conducted at 1, 3, 5, 7, 9, and 11 d postpartum. Postpartum diseases were recorded and defined as follows: hypocalcemia was diagnosed based on low colostrum production, lethargy, and muscle tremors, and treated with a calcium bolus as part of the farm's management practice. Mastitis was diagnosed by the California Mastitis Test and confirmed by laboratory analysis. A retained placenta was diagnosed by the retention of fetal membranes for >24 h following parturition. Ketosis was diagnosed when circulating BHB was >2.1 mmol/L, measured using a Precision Xtra Analyzer (Abbott). Cows with ketosis were treated with propylene glycol for 3 d as part of farm protocol. Cows experiencing disease were examined by the veterinarian and treated by the farm management.

Blood Metabolic Profiling, Analyses, and Descriptive Thresholds

Blood samples were obtained 10 d before the expected parturition date (8.56 ± 2.8 d to parturition) and 7 d postpartum from the vein or artery in the intercoccygeal space before morning feeding and before adipose tissue collection using a 21-gauge vacutainer needle (cat. no. 368650; BD Vacutainer, Becton Dickinson) for metabolic profiling. After blood collection, tubes with lithium heparin were placed on ice, and tubes with clot activator were kept at room temperature (21°C) for approximately 1.5 h until centrifugation. Serum and plasma were obtained by centrifugation at $1,300 \times g$ for 15 min at 21°C and 4°C, respectively, and aliquots were frozen at -80°C until further analysis. Serum concentrations of NEFA (cat. no. 99934691; FUJIFILM Medical Systems Valhalla, NY), and plasma concentrations of glucose (cat. no. 99703001; FUJIFILM Medical Systems), BHB (cat. no. 700190; Cayman Chemical; Ann Arbor, MI), insulin (Bovine Insulin ELISA; Mercodia; cat. no. 10120101; Uppsala, Sweden), and glutathione peroxidase activity (GPx, EC 1.11.9; cat. no. 703102; Cayman Chemical) were analyzed by commercial kits. Plasma was also analyzed for the concentrations of the following biomarkers on the ILAB 650 clinical autoanalyzer (Instrumentation Laboratory, USA S.p.a., Werfen Co., Milan, Italy): calcium, albumin, globulin, haptoglobin, total bilirubin, and the activity of γ -glutamyltransferase (GGT, EC 2.3.2.2), and paraoxonase (PON, EC 3.1.1.2; Bionaz et al., 2007; Trevisi et al., 2012; Mezzetti et al., 2019). This study was performed on a subset of animals reported by Lima et al. (2025); therefore, intra- and interassay CV (%) for indicators measured by commercial kits are reported for the entire original set of animals. The respective intra- and interassay CV% were: glucose 4.6% and 4.2%, GPx

4.0% and 8.1%, NEFA 5.2% and 4.2%, BHB 3.7% and 8.9%, and insulin 7.1% and 9.3%.

For further correlation analyses, blood NEFA concentrations were categorized by period, according to thresholds established by Ospina et al. (2010): In PRE, NEFA < 0.3 mmol/L was considered optimal (OL, and NEFA $\geq$ 0.3 mmol/L was considered elevated lipolysis (EL). In POST, NEFA < 0.6 mmol/L was considered OL, and NEFA $\geq$ 0.6 mmol/L was considered EL. Postpartum disease development was used as a categorical variable based on whether a cow did (Yes, Y) or did not (No, N) develop ketosis, retained placenta, mastitis, or hypocalcemia in the first 7 DIM.

Adipose Tissue Collection

Subcutaneous adipose tissue samples from 16 dairy cows were collected by blunt dissection from alternate sides of the tailhead region at -10 (PRE, 8.56 ± 2.8 d to parturition) and 10 d (POST) relative to parturition. Briefly, the biopsy site was shaved and disinfected with alcohol, followed by iodine. Local anesthesia was administered by subcutaneous injection of 15 mL of 2% lidocaine (cat. no. 13985-222-05; Vet One, Boise, ID). A 3-cm skin incision on the tubera ischii bone area was made using a scalpel attached to a no. 10 blade. Approximately 500 mg of SAT were collected, and the incision was closed with surgical skin staples. Vetasan ointment (chlorhexidine gluconate 4%; cat. no. 55001-1; Kinetic VET; Lexington, KY) was applied over staples to prevent infection. Tissue samples were rinsed with saline solution, flash-frozen in liquid nitrogen, and stored at -80°C until further analysis. Health indicators were monitored for 3 d after each procedure, and surgical skin staples were removed 5 d after biopsy.

RNA Extraction, Quantitative Real-Time PCR, and Analysis

To quantify the expression of target genes, RNA was extracted from matched PRE and POST SAT samples ($n = 16$ cows; 32 samples), as previously described by Fiallo Diez et al. (2024), with slight modifications. Briefly, 70 to 80 mg of each SAT sample were placed in a 2-mL RNase-free O-ring tube with four 3.0-mm zirconium beads (cat. no. D1132-30TP; Benchmark Scientific Inc.) and 1 mL of cold Trizol. Tissue was homogenized twice for 2.5 min using a BeadBlaster (Benchmark Scientific Inc., Sayreville, NJ), with a 2-min interval between homogenization periods to cool the samples on ice and prevent RNA degradation. Following homogenization, the homogenate was transferred to a new 2-mL RNase-free microcentrifuge tube (cat. no. 14-666-313; Fisher Scientific) with 200 μ L of phenol:chloroform (cat. no.

Table 1. TaqMan gene expression assays from Life Technologies (Invitrogen) used in quantitative real-time PCR (RT-qPCR)

Gene name	Gene symbol	Gene network	Assay identification
Peroxisome proliferator-activated receptor gamma	<i>PPARG</i>	Adipogenesis	Bt03217547_m1
Perlipin-1	<i>PLIN1</i>	Adipogenesis	Bt03257414_m1
Fatty acid synthase (EC 2.3.1.85)	<i>FASN</i>	Adipogenesis	Bt0310481_m1
Hormone-sensitive lipase (EC 3.1.1.79)	<i>LIPE</i>	Adipogenesis	Bt0329341_s1
Adiponectin	<i>ADIPOQ</i>	Adipogenesis	Bt03211909_m1
Leptin	<i>LEP</i>	Adipogenesis	Bt03211909_m1
Diglycerol-acyltransferase 1 (EC 2.3.1.20)	<i>DGAT1</i>	Adipogenesis	Bt03251718_g1
Myelin protein zero	<i>MPZ</i>	Nervous system	Bt03253533_m1
Tyrosine hydroxylase (EC 1.14.14.36)	<i>TH</i>	Nervous system	Bt03211746_m1
Dopamine β -hydroxylase (EC 1.14.17.1)	<i>DBH</i>	Nervous system	Bt03217365_m1
β -Tubulin 3	<i>TUBB3</i>	Nervous system	Bt03237041_m1
Nerve growth factor receptor	<i>NGFR</i>	Nervous system	Bt03217725_m1
SRY-Box 10	<i>SOX10</i>	Nervous system	Bt04316887_m1
S100 calcium binding protein B	<i>S100B</i>	Nervous system	Bt03226845_m1
PPARG coactivator 1 α	<i>PGC1A</i>	Thermogenesis	Bt01208835_m1
Early B-cell factor 2	<i>EBF2</i>	Thermogenesis	Bt03251011_m1
Transmembrane protein 26	<i>TMEM26</i>	Thermogenesis	Bt03277907_m1
Uncoupling protein 1	<i>UCP1</i>	Thermogenesis	Bt04301238_m1
β 2 Microglobulin	<i>B2M</i>	Housekeeper	Bt03251628_m1
Eukaryotic initiation factor 3 subunit K	<i>EIF3K</i>	Housekeeper	Bt03226565_m1

AM9732; Invitrogen) and shaken for 15 s to promote phase separation. After centrifugation at $12,000 \times g$ for 15 min at 4°C , the clear aqueous layer was transferred to a new microcentrifuge tube containing an equal volume of 70% ethanol (cat. no. BP2818100; Fisher Bioreagents) and vortexed. Using the RNeasy mini kit (RNeasy mini kit, cat. no. 74104; QIAGEN), the solution was centrifuged at $10,000 \times g$ for 15 s at room temperature ($\sim 25^{\circ}\text{C}$) through spin columns and washed 2 times with RPE buffer. Total RNA was eluted with 100 μL of RNase-free water. RNA purity was assessed by 260/280 ratio of absorbance using the Take3 microvolume plate on a Cytation5 multimode plate reader with values falling between 1.8 and 2.0 (cat. no. TAKE3-SN; Agilent-BioTek). RNA integrity was evaluated in a representative subset of samples ($n = 5$ per period) using the Agilent 4200 TapeStation (Agilent Technologies). RNA integrity numbers (RIN) were low in both PRE (2.9 ± 1.0) and POST (2.1 ± 0.5) samples, which is consistent with the high lipid content and known susceptibility of adipose tissue RNA to degradation. Importantly, RIN values did not differ between periods (Mann–Whitney test; $P = 0.21$), indicating comparable RNA integrity across experimental groups. Using 200 ng of initial RNA, cDNA was synthesized via a High-Capacity cDNA Reverse Transcription Kit (cat. no. 4368814; ThermoScientific) in a MiniAmpPlus thermal cycler (cat. no. A37835; Applied Biosystems, Waltham, MA). Quantitative real-time PCR (RT-qPCR) was performed to assess gene expression, with 2 technical replicates per gene for each cow tissue sample using a QuantStudio 6 Pro System (cat. no. A43180; Applied Biosystems, Waltham, MA) using TaqMan gene expression assays and reagents (Life Technologies Inc., Carlsbad, CA; Table 1). β 2-microglobulin

(*B2M*) and eukaryotic initiation factor 3 subunit K (*EIF3K*) were selected as housekeeping genes because they showed consistent expression across the 2 periods (PRE vs. POST). Expression data were normalized to the geometric mean of housekeeping gene expression, as described by Ford et al. (2024).

Protein Extraction, Sample Cleaning, and Western Blot

Western Blot analysis of our SAT samples allows for the detection of axonal proteins because nearly all peripheral neuron somas innervate from the spinal cord via the dorsal root ganglia (Servin-Vences et al., 2023; Wolfson et al., 2023) or paravertebral ganglia (Cardoso et al., 2021; Huesing et al., 2021), and the vagal fibers from the brain (Chang et al., 2015; Prescott and Liberles, 2022; Zhao et al., 2022); thus, RNA changes from neurons would be undetectable. Total protein was extracted from 350 mg of SAT collected from each cow at both PRE and POST time points ($n = 16$ cows; 32 total samples). Briefly, each sample was homogenized using a hand-held homogenizer (cat. no. 10032–330; VWR) in 500 μL of radioimmunoprecipitation assay buffer (cat. no. J63306; ThermoScientific) supplemented with 10 μL of phosphatase inhibitor cocktail II (cat. no. J61022.AA; ThermoScientific) and one cComplete mini, EDTA-free protease inhibitor cocktail (cat. no. 11836170001; SigmaAldrich) tablet per 10 mL of buffer. Protein concentration was determined by using a Pierce bicinchoninic protein assay kit (cat. no. 23227; ThermoScientific).

Before electrophoresis, protein isolates underwent lipid cleanup using an acetone procedure as described by Park et al. (2019). Briefly, 40 μg or 20 μg of protein

isolate was incubated in 4 volumes of 100% acetone at -20°C for 1 h. Next, samples were centrifuged at $15,000 \times g$ for 10 min at 4°C , the supernatant was removed, and the protein isolates were washed by adding 80% acetone and centrifuging at $15,000 \times g$ for 5 min at 4°C . After supernatant removal, isolates were allowed to dry for 20 min at room temperature. Once dry, protein samples were resuspended in solution by adding 4 $\times$ sample buffer (cat. no. LC2570; ThermoScientific), 10 $\times$ reducing buffer (cat. no. B0009; ThermoScientific), and deionized water, then vortexed at 25,000 rpm for 3 min at room temperature or until the protein pellet was visibly dissolved. Immediately after resuspension, isolates were incubated at 95°C for 5 min and then placed on ice. Bolt 4%–12% Bis-Tris Plus gels (cat. nos. NW04120BOX and NW04125BOX; Invitrogen) were used to separate the protein isolates at room temperature using 2-(N-morpholino)ethanesulfonic acid buffer for 24 min at a constant 200 V. We used the prestained SeeBlue Plus2 (cat. no. LC5925; ThermoScientific) or the iBright (cat. no. LC5615; ThermoScientific) as protein size markers. Gels were transferred by semidry transfer using the Power Blotter system (cat. no. PB0012; ThermoScientific).

For detection of targeted proteins, blots were first blocked in a Tris-buffered saline + Tween-20 (TBST) blocking solution containing 0.05% Tween 20 (cat. no. 85113; Thermo-Scientific) in Tris-buffered saline (TBS, cat. no. J75892-K8; ThermoScientific) with 5% bovine serum albumin (cat. no. BP1600-1, Thermo-Scientific) for 1 h at room temperature (27°C). Following blocking, blots were incubated with the primary antibodies described in Table 2 in blocking solution at 4°C overnight. Blots were then washed with 0.05% TBST solution without BSA and incubated with secondary antibodies (Table 2) at room temperature (18°C – 21°C) for 1 h in the blocking solution. Horseradish peroxidase–conjugated antibodies were activated using the SuperSignal West Pico PLUS Chemiluminescent Substrate kit (cat. no. 34580; ThermoScientific) according to the manufacturer's instructions and imaged and read for band density on the iBright 1500 (ThermoFisher Scientific). For normalization, total protein was calculated using a Pierce Reversible Protein Kit (cat. no. 24580; ThermoScientific; Supplemental Figure S1A, see Notes) following the manufacturer's instructions after all targeted proteins had been detected. Target proteins were normalized by dividing the local background-corrected density of the target protein by the local background-corrected density of total protein in each respective blot lane to control for loading and transfer variability. All protein data are shown as normalized values. Activation of catecholamine production was assessed by a ratio of phospho-TH Serine 40 residues (pTHser40) over total tyrosine hydroxylase (TH) protein (pTHser40:TH). Activation of hormone-sensitive lipase

(HSL) was assessed by a ratio of phospho-HSL Ser660 residue (pHSLSer660) or phospho-HSL Ser563 residue (pHSLSer563) over total HSL protein (pHSLSer660: HSL or pHSLSer563: HSL).

Statistical Analyses

Statistical analyses were performed using PROC MIXED in SAS (version 9.4; SAS Institute Inc., Cary, NC). A repeated-measures mixed-effects model was used in the main outcomes of the study. Period (PRE vs. POST) was treated as the repeated measure within cow, and cow was nested within parity to account for individual animal effects. Fixed effects included parity, period, and the parity $\times$ period interaction. Parity was categorized as cows entering their second lactation ($n = 4$), third lactation ($n = 7$), or fourth and greater lactation ($n = 5$). The period was defined as PRE or POST. The previous 305-d milk was initially included as a covariate but excluded from the final model based on improved fit (lower Akaike information criterion [AIC] or Bayesian information criterion [BIC]). Covariance structures (unstructured and compound symmetry) were evaluated, and compound symmetry was selected based on lower AIC and BIC values.

Spearman correlations were performed using PROC CORR in SAS to assess associations between adipose tissue gene expression or protein abundance and circulating metabolites, inflammatory markers, and animal-level variables (BCS difference, parity, previous milk production, and disease development). Correlations with $P \leq 0.10$ were considered for further visualization and were graphically assessed using a scatter plot in GraphPad Prism v.10. For grouped comparisons derived from continuous variables, Wilcoxon signed-rank tests were performed.

RESULTS AND DISCUSSION

Systemic Metabolic Adaptation During the Periparturient Period

The descriptive effects of period, parity, and their interaction on BCS and circulating metabolic and inflammatory markers are shown in Table 3. Differences in circulating metabolites were driven primarily by period effect rather than parity, with only albumin ($P = 0.08$) and total bilirubin ($P = 0.03$) increasing in cows in third lactation, with no difference between cows in their second or fourth or greater lactation. Overall, animals demonstrated typical transition-period changes, with decreases in BCS ($P = 0.007$), and increases in serum NEFA ($P = 0.0015$) and BHB ($P < 0.0015$), and a decrease in

Table 2. Antibodies used for western blotting of adipose tissue lysates

Target protein name ¹	Protein symbol	Host species	Concentration used	Manufacturer	Catalog no.
Primary antibody	—	—	—	—	—
Adrenergic receptor $\beta 2$	ADRB2	Rabbit	1:500	Invitrogen	PA5–14117
Adrenergic receptor $\beta 3$	ADRB3	Rabbit	1:1,000	Invitrogen	PA5–97131
Calcitonin gene-related peptide	CGRP	Rabbit	1:1,000	Invitrogen	MA5–44524
Dopamine β -hydroxylase	DBH	Sheep	1:5,000	Invitrogen	PA3–925
Early B-cell factor 2	EBF2	Sheep	1:400	Invitrogen	PA5–47890
Glial fibrillary acidic protein	GFAP	Rabbit	1:1,000	Invitrogen	PA5–16291
Hormone-sensitive lipase	HSL	Rabbit	1:2,000	Invitrogen	PA5–20298
Phospho-HSL Ser563 residue	pHSL _{Ser563}	Rabbit	1:2,000	Invitrogen	PA5–104600
Phospho-HSL Ser660 residue	pHSL _{Ser660}	Rabbit	1:1,000	Cell Signaling	45804S
Myelin protein zero	MPZ	Rabbit	1:1,000	Invitrogen	PA5–37179
Nerve growth factor receptor	NGFR/p75NTR	Mouse	1:500	Invitrogen	14–9400–80
PR domain-containing 16	PRDM16	Rabbit	1:1,000	Invitrogen	PA5–20872
Protein gene product 9.5	PGP9.5/UCHL1	Mouse	1:5,000	Invitrogen	480012
Perilipin 1	PLIN1	Mouse	1:1,000	Invitrogen	MA5–27861
S100 calcium binding protein B	S100B	Mouse	1:500	Invitrogen	PA5–78161
Norepinephrine transporter	NET	Mouse	1:1,000	Invitrogen	PA5–78161
Transmembrane protein 26	TMEM26	Rabbit	1:1,000	Invitrogen	PA5–23477
β -Tubulin 3	TUBB3	Mouse	1:1,000	Invitrogen	MA1–118X
Tyrosine hydroxylase	TH	Rabbit	1:1,000	Invitrogen	PA5–85356
Phospho-TH Ser40 residue	pTH _{Ser40}	Rabbit	1:1,000	Invitrogen	PA5–104764
Secondary antibody	—	—	—	—	—
HRP conjugated goat anti-rabbit	N/A ²	Goat	1:10,000	Invitrogen	31460
HRP conjugated goat anti-mouse	N/A	Goat	1:10,000	Invitrogen	A28177
HRP conjugated donkey anti-sheep	N/A	Donkey	1:10,000	Invitrogen	A16041

¹HRP = horseradish peroxidase.²N/A = not applicable.

insulin ($P < 0.0001$) in POST relative to PRE. These changes are expected metabolic adaptations to the negative energy balance occurring during the periparturient period in dairy cows, as previously reviewed (Drackley, 1999). Of note, in PRE, we observed optimal lipolysis in 94% (15/16) of cows, compared with only 43% (7/16) in POST, indicating an excessive lipolytic response in early lactation. Accordingly, among the 16 cows included in this study, 7 developed diseases within the first 7 DIM, including hypocalcemia ($n = 1$), mastitis ($n = 2$), retained placenta ($n = 2$), and ketosis ($n = 3$). Given the low disease prevalence and small sample size, we did not assess the effect of parity on disease development.

Liver function, inflammatory status, and oxidative balance were evaluated using glutathione peroxidase (GPx) activity, γ -glutamyltransferase (GGT) activity, and paraoxonase activity, as well as plasma concentrations of calcium, albumin, globulin, total bilirubin, and haptoglobin (Table 3). There were no differences between PRE and POST periods for GPx ($P = 0.49$), calcium ($P = 0.25$), GGT ($P = 0.97$), or haptoglobin ($P = 0.24$). Concentrations of albumin ($P = 0.014$) and globulin ($P = 0.031$) as well as paraoxonase activity ($P = 0.04$) decreased, whereas total bilirubin ($P < 0.0001$) increased in POST versus PRE. Altogether, these changes are consistent with hepatic adaptation and activation of an acute-phase response in early postpartum cows, as previously described (Trevisi et al., 2012; Drackley et al., 2025). Although markers of

oxidative balance did not differ between periods based on GPx and GGT activity, additional indices would be required to fully assess oxidative stress status (Table 3).

Parity Is Associated with Sensory Neural Markers in Tailhead SAT

The nervous system regulates adipose tissue lipolysis, adipocyte quantity, satiety, and thermogenesis activation (Bartness and Song, 2007; Murphy et al., 2013; Ikeda and Yamada, 2020). Thus, in this study, we assessed changes in markers of local adipose tissue innervation in multiparous periparturient dairy cows. Although the adrenergic system of the SNS garners most of the attention in its regulation of adipose tissue lipolysis and thermogenesis, the sensory system plays a key role as a regulator of fat mass perception and inflammation, as primarily reported in rodent white adipose tissue (Fishman and Dark, 1987; Murphy et al., 2013; Dumont et al., 2025).

To evaluate sensory innervation, we quantified calcitonin gene-related peptide (CGRP) protein abundance in SAT. CGRP was detected predominantly in its propeptide form (ProCGRP), as confirmed using bovine thalamus, spinal cord, and cortex as positive control (Supplemental Figure S1B). ProCGRP abundance did not differ between the PRE and POST periods (Figure 1A–B; $P = 0.99$), indicating stability of sensory innervation across the periparturient transition. In contrast, a significant parity

Table 3. Least squares means of BCS and circulating blood metabolites during the PRE period (-8.6 ± 2.3 d relative to parturition) and POST period (7 d after parturition) in multiparous dairy cows (3.5 ± 1.6 lactations; $n = 16$) in their second ($n = 4$), third ($n = 7$), and fourth or greater ($n = 5$) lactations¹

Indicator	Period				Lactation number					Lactation × period, <i>P</i> -value
	PRE	POST	SEM	<i>P</i> -value	2nd	3rd	≥4th	SEM	<i>P</i> -value	
BCS	3.50	3.18	0.08	0.007	3.16	3.52	3.34	0.12	0.098	0.856
NEFA (mmol/L)	0.23	0.95	0.12	0.002	0.44	0.77	0.57	0.15	0.250	0.807
Glucose (mmol/L)	3.93	3.59	0.19	0.136	3.80	3.70	3.77	0.29	0.954	0.656
BHB (mmol/L)	0.18	0.83	0.11	0.002	0.38	0.62	0.52	0.16	0.500	0.344
Insulin (μIU/mL)	10.45	1.66	0.66	<0.0001	5.07	6.36	6.74	1.01	0.463	0.402
Glutathione peroxidase (U/mL)	141.16	156.68	15.62	0.491	140.38	154.90	151.48	21.72	0.866	0.367
Calcium (mmol/L)	2.59	2.48	0.06	0.247	2.57	2.61	2.44	0.08	0.206	0.206
Albumin (g/L)	35.99	33.44	0.64	0.014	34.54	36.08	33.51	0.88	0.075	0.053
Globulin (g/L)	35.41	31.82	1.22	0.031	31.51	32.79	36.54	1.89	0.140	0.146
Haptoglobin (g/L)	0.22	0.33	0.07	0.242	0.22	0.26	0.33	0.10	0.792	0.243
Total bilirubin (μmol/L)	0.55	4.96	0.49	<0.0001	1.46	4.07	2.73	0.68	0.027	0.025
γ-Glutamyltransferase (U/L)	20.05	20.00	0.98	0.965	20.32	20.18	19.58	1.56	0.926	0.445
Paraoxonase (U/mL)	79.96	72.72	3.45	0.037	81.34	80.79	66.90	6.00	0.132	0.000

¹Additional information is provided in Supplemental Table S1 (see Notes).

effect was observed, with cows in their third lactation exhibiting greater ProCGRP abundance compared with cows in their second or fourth or greater lactation (Figure 1C; $P = 0.006$). This parity-specific pattern suggests enhanced sensory neural signaling in midparity cows, which may reflect differences in adipose tissue sensing or inflammatory feedback across successive lactations, independent of acute periparturient changes, and support further evaluation of additional components of adipose tissue neural regulation.

Periparturient Changes in Schwann Cell Markers Suggest Altered Myelination

Schwann cells support peripheral nerve transmission by ensheathing axons and forming layers of myelin, a process critical for neuronal function (Mirsky et al., 2008; Willows et al., 2023). Myelin protein zero (MPZ) is the main myelination protein of the peripheral nervous system and is secreted by myelinating Schwann cells, which specifically express S100B (Mata et al., 1990). Accordingly, MPZ and S100B were used as markers of myelination and myelinating Schwann cells in the present study.

Gene expression of *S100B* decreased from PRE to POST (Figure 1D; $P < 0.0001$), whereas *MPZ* expression did not differ between periods (Figure 1D; $P = 0.13$). To assess changes in Schwann cell populations more broadly, nerve growth factor receptor (*NGFR*) and SRY-Box 10 (*SOX10*) were evaluated as Schwann cell markers (Mirsky et al., 2008; Reiprich et al., 2010). *NGFR* expression increased from PRE to POST (Figure 1D; $P = 0.0002$), and *SOX10* expression remained unchanged (Figure 1D; $P = 0.35$).

To determine whether observed changes reflected alterations in Schwann cell phenotype rather than their

abundance, *MPZ*, *S100B*, and *NGFR* expression were normalized to *SOX10*. Following normalization, *MPZ* and *S100B* expression decreased from PRE to POST (Supplemental Figure S1C–D; $P = 0.02$ for both), whereas *NGFR* expression remained unchanged (Supplemental Figure S1E; $P = 0.13$). These results indicate that periparturient changes primarily involve reduced myelination-associated gene expression rather than changes in overall Schwann cell number or *NGFR* expression.

In contrast to myelinating Schwann cells, nonmyelinating Schwann cells support axonal maintenance, neuronal survival, and nerve regeneration and are typically organized in Remak bundles or maintained in a responsive state during metabolic or immunological insults (Griffin and Thompson, 2008). The coordinated reduction in *SOX10*-normalized *MPZ* and *S100B* gene expression, coupled with stable *NGFR* expression, indicates a potential shift in Schwann cell phenotype from a myelinating toward a nonmyelinating state, suggesting neural remodeling within adipose tissue of periparturient cows (Lovatt et al., 2022).

Protein-Level Validation Reveals Limited Period Effects but Parity-Dependent Neural Remodeling

To validate gene expression analysis, we performed western blotting of MPZ, S100B, NGFR, and glial fibrillary acidic protein (GFAP), a marker of nonmyelinating Schwann cells (Yang and Wang, 2015), in adipose tissue protein lysates. At the protein level, only modest period-dependent differences were observed. MPZ protein abundance tended to decrease from PRE to POST (Figure 1A–B; $P = 0.06$), whereas S100B protein abundance remained unchanged between periods (Figure 1A–B; $P = 0.25$). This divergence suggests post-transcriptional or posttranslational regulations of S100B.

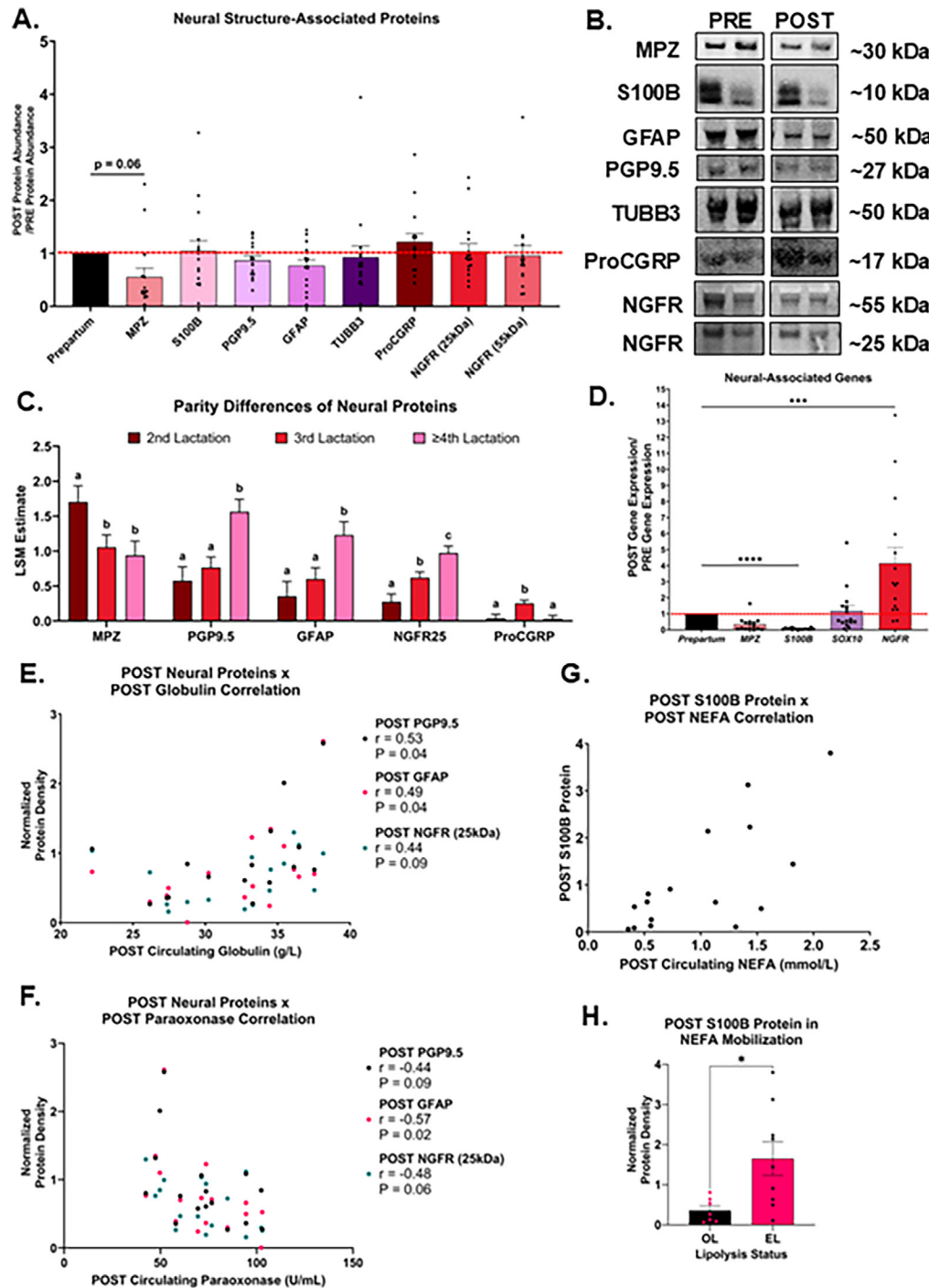


Figure 1. Changes of selected peripheral nervous system markers in the SAT of the PRE period (8.56 ± 2.3 d to parturition; mean $\pm$ SD; $n = 16$) and the postpartum period ($+7$ d after parturition, $n = 16$) of multiparous dairy cows (3.5 ± 1.6 lactations, $n = 16$). (A) Neural-structure associated protein abundance of MPZ, S100B, PGP9.5, GFAP, TUBB3, ProCGRP, NGFR_{25kDa}, and NGFR_{55kDa}. (B) Representative western blots of MPZ, S100B, glial fibrillary acidic protein (GFAP), protein gene product 9.5 (PGP9.5), tubulin- $\beta 3$ (TUBB3), procalcitonin gene-related product (ProCGRP), and immunoreactive band of NGFR (55 and 25 kDa). (C) Parity differences of neural proteins MPZ, PGP9.5, GFAP, NGFR25, and ProCGRP. (D) Neural-associated gene markers myelin protein zero (MPZ), S100 calcium binding protein B (S100B), SRY-BOX-10 (SOX10), and nerve growth factor receptor (NGFR). (E, F) Visualization of Spearman correlation between postpartum PGP9.5, GFAP, and NGFR_{25kDa} (POST neural proteins) with circulating globulin (E) or paraoxonase (F). (G) Visualization of Spearman correlation between postpartum S100B with circulating NEFA. (H) Abundance of S100B protein by lipolysis status. Protein abundance and gene expression analyzed by a repeated-measures mixed-effects model with the repetition of period by cow nested in parity, and parity, period, and the interaction of parity and period as fixed effects. Presented as fold-change of postpartum measurement and prepartum measurement. Error bars indicate SEM. Different letters signify group differences (significance threshold is 0.05). * $P \leq 0.05$, *** $P < 0.001$, **** $P < 0.0001$.

NGFR protein abundance was also unchanged between periods; however, 2 distinct bands were detected (Figure 1B). The ~55 kDa band likely represents the mature glycosylated or partially glycosylated NGFR (NGFR_{55kDa}; Figure 1A–B; $P = 0.69$), while the ~25kDa band corresponds to the reported intracellular domain of NGFR (NGFR_{25kDa}; Figure 1A–B; $P = 0.79$; Ceni et al., 2010). GFAP protein abundance likewise did not differ from PRE to POST (Figure 1A–B; $P = 0.31$), consistent with stable *SOX10* gene expression. Collectively, these findings indicate that the periparturient period is primarily characterized by modest alterations in myelination-related proteins rather than large-scale changes in Schwann cell abundance or neural density.

To determine whether neural remodeling extended beyond myelination, we quantified the pan-neuronal markers β -tubulin III (TUBB3) and protein gene product 9.5 (PGP9.5), also known as ubiquitin carboxyl-terminal hydrolase L1 (UCHL1). Neither TUBB3 (Figure 1A–B; $P = 0.17$) nor PGP9.5 (Figure 1A–B; $P = 0.19$) differed from PRE to POST, indicating that overall neural abundance was stable across the transition period. In contrast, parity exerted significant effects on neural-associated proteins in adipose tissue. Abundance of PGP9.5/UCHL1 increased with parity (Figure 1C; $P = 0.002$) and GFAP exhibited a similar upward trend (Figure 1C; $P = 0.02$), but MPZ abundance tended to decrease with parity (Figure 1C; $P = 0.06$). Together, these results suggest that older cows exhibit increased adipose tissue innervation accompanied by reduced myelination, consistent with parity-dependent neural remodeling.

Neural Markers Are Associated with Inflammation and Lipolysis in Early Lactation

As a descriptive analysis, associations between metabolic and inflammatory markers and selected neural markers were evaluated during the POST period. Positive associations were revealed between globulin concentrations and protein abundance of PGP9.5/UCHL1, GFAP, and NGFR_{25kDa} (Figure 1E; $r = 0.53$, $P = 0.04$; $r = 0.49$, $P = 0.05$; $r = 0.44$, $P = 0.09$, respectively). In contrast, paraoxonase activity was negatively associated with these neural markers (Figure 1F; $r = -0.44$, $P = 0.09$; $r = -0.57$, $P = 0.02$; $r = -0.48$, $P = 0.06$, respectively). Because circulating globulin increases and paraoxonase activity decreases during systemic inflammation (Bertoni and Trevisi, 2013), these relationships suggest an inflammation-linked neural response in adipose tissue during early lactation. Given the established roles of PGP9.5/UCHL1, GFAP, and NGFR in neural remodeling and plasticity (Triolo et al., 2006; Cao et al., 2018; Blaszkiewicz et al., 2019), our findings support the presence

of active neural remodeling in the context of postpartum inflammatory adaptation.

In rodent models, S100B has been extensively characterized as a bidirectional proinflammatory adipokine, mediator of neurite outgrowth, and contributor to beige adipose tissue biogenesis macrophage-adipocyte communication (Fujiya et al., 2014; Zeng et al., 2019, Huang et al., 2023). Our study demonstrated that the abundance of S100B in POST was positively correlated with circulating NEFA concentrations (Figure 1G; $r = 0.63$, $P = 0.009$), suggesting an association between S100B protein expression and heightened lipolysis (Kosteli et al., 2010; Sun et al., 2011; Contreras et al., 2017). Consistent with this observation, cows classified as having EL in POST (> 0.6 mmol/L NEFA) had higher S100B protein in adipose tissue compared with the group of cows with OL NEFA concentrations (< 0.6 mmol/L NEFA; Figure 1H; $P = 0.02$; Ospina et al., 2010). These findings link increased S100B abundance with EL and inflammation accompanying adipose tissue mobilization, potentially mediated by adipocyte-macrophage crosstalk.

Despite its reported role in beige adipose tissue development, S100B abundance was not associated with expression of canonical beiging markers, including B-cell factor 2 (EBF2), transmembrane protein 26 (TMEM26), and PR domain-containing 16 (PRDM16; Supplemental Figure S1F). This suggests that in periparturient dairy cows adipose tissue expression of S100B is more closely linked to inflammation and lipolytic activity than induction of beiging.

Jointly, these results indicate that postpartum inflammation and lipolysis are associated with coordinated changes in adipose tissue neural markers. When considered alongside parity-dependent increases in neural abundance and concurrent reductions in myelination, these findings support a model in which adipose tissue neural remodeling contributes to metabolic and inflammatory adaptation during the transition from pregnancy to early lactation.

Catecholamine Biosynthesis Markers Are Stable Across the Periparturient Transition

Catecholamines regulate lipolysis, hyperplasia, non-shivering thermogenesis, beiging, and glucose uptake in adipose tissue through the number of their receptors and the concentration of catecholamines released (Jaster and Wegner, 1981; Evans et al., 2019). We assessed the production of catecholamines in SAT by quantifying the gene expression of the rate-limiting enzyme, tyrosine hydroxylase (*TH*), responsible for the hydroxylation of L-tyrosine to the dopamine precursor, L-3,4-dihydroxyphenylalanine (Figure 2A).

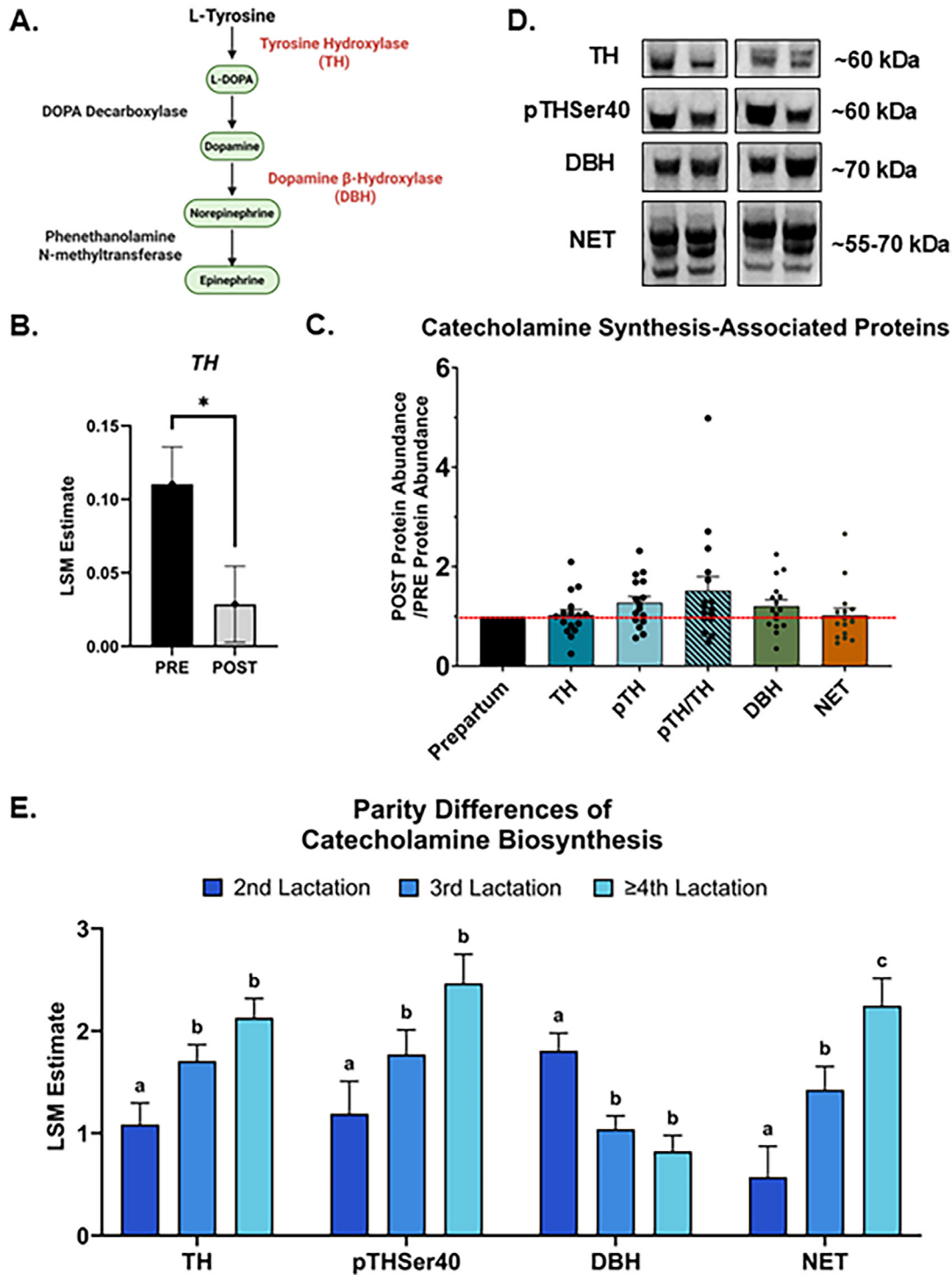


Figure 2. Effects on selected enzymes and transporters within the catecholamine biosynthesis pathways by parturition in the SAT of the PRE period (8.56 ± 2.3 d to parturition, mean $\pm$ SD, $n = 16$) and the POST period (+7 d after parturition, $n = 16$) of multiparous dairy cows (3.5 ± 1.6 lactations, $n = 16$). (A) Schematic diagram of catecholamine biosynthesis pathway with regulatory enzymes evaluated in this study in red. (B) Gene expression of tyrosine hydroxylase (TH). (C) Catecholamine synthesis-associated proteins TH, phospho-TH Serine 40 residues (pTH_{Ser40}), the activation ratio of pTH_{Ser40}:total TH, dopamine β -hydroxylase (DBH), and norepinephrine transporter (NET) protein abundance in reference to prepartum abundance. (D) Representative western blots of TH, pTH_{Ser40}, DBH, and NET. (E) Parity differences in protein abundance of TH, pTH_{Ser40}, DBH, and NET. Protein abundance and gene expression were analyzed by a repeated-measures mixed-effects model with the repetition of period by cow nested in parity, and parity, period, and the interaction of parity and period as fixed effects and presented as fold-change of postpartum measurement and prepartum measurement. Error bars indicate SEM. Different letters signify group differences (significance threshold is 0.05). * $P < 0.05$. Panel A created in BioRender. Strieder-Barboza, C. (2026).

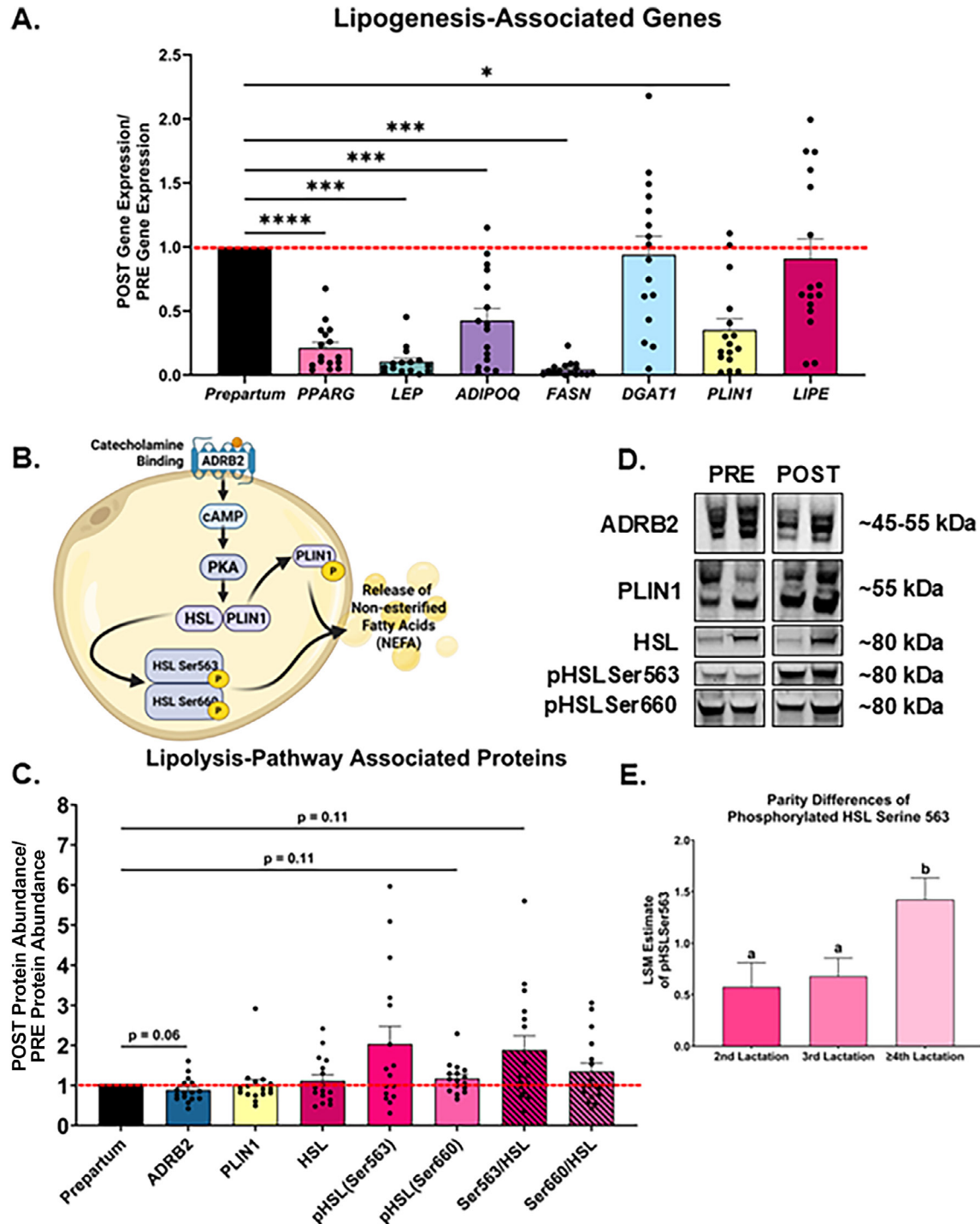


Figure 3. Effects on the catecholamine-mediated lipolysis pathways by parturition in the SAT of the PRE period (8.56 ± 2.3 d to parturition, mean $\pm$ SD, $n = 16$) and the POST period ($+7$ d after parturition, $n = 16$) multiparous dairy cows (3.5 ± 1.6 lactations, $n = 16$). (A) Lipogenesis-associated genes leptin (*LEP*), adiponectin (*ADIPOQ*), fatty acid synthase (*FASN*), diglycerol-acyltransferase 1 (*DGAT1*), perilipin 1 (*PLIN1*), and HSL (*LIPE*). (B) Schematic diagram of HSL release of nonesterified fatty acids (NEFA). (C) Lipolysis-associated protein abundance of adrenergic receptor β_2 (ADRB2), PLIN1, HSL, pHSLSer563, pHSLSer660, the activation ratio of pHSLSer563:total HSL, and the activation ratio of pHSLSer660:total HSL. (D) Representative western blots of ADRB2, perilipin-1 (PLIN1), HSL, phospho-HSL serine 563 residues (pHSLSer563), and phospho-HSL serine 660 residues (pHSLSer660). (E) Parity differences in the protein abundance of pHSLSer563. Protein abundance and gene expression were analyzed by a repeated-measures mixed-effects model with the repetition of period by cow nested in parity, and parity, period, and the interaction of parity and period as fixed effects. Presented as fold-change of postpartum measurement and prepartum measurement. Error bars indicate SEM. Different letters signify group differences (significance threshold is 0.05). * $P \leq 0.05$, *** $P < 0.001$, **** $P < 0.0001$. Biological schematic created in BioRender. Strieder-Barboza, C. (2026).

Adipose gene expression of *TH* decreased from PRE to POST (Figure 2B; $P = 0.05$), and total TH protein abundance remained the same (Figure 2C–D; $P = 0.59$). Additionally, we observed no period effect on the abundance of phosphorylated TH at its activation site, serine 40 (pTH_{Ser40}, Figure 2C–D; $P = 0.27$) or the activation ratio of pTH_{Ser40}:TH (Figure 2C–D; $P = 0.30$) across the periparturient transition.

To further investigate norepinephrine (NE) biosynthesis by SAT, we assessed gene and protein expression of dopamine β -hydroxylase (*DBH*), the enzyme that converts dopamine to NE (Molinoff and Axelrod, 1971). Notably, *DBH* transcripts were detected in fewer than 5% of SAT samples in both PRE and POST, precluding statistical analysis. In contrast, DBH protein was abundantly detected and expressed at comparable levels in PRE and POST (Figure 2C–D; $P = 0.83$). Although non-neuronal catecholamine synthesis has been reported in murine adipocytes (Bartness et al., 2010a), the low abundance of *DBH* transcripts observed here suggests that catecholamine production within periparturient bovine SAT is minimal outside of neural sources.

Beyond biosynthesis, catecholamine availability is regulated by clearance mechanisms such as reuptake via the norepinephrine transporter (NET) located at presynaptic terminals (Axelrod and Kopin, 1969; Mandela and Ordway, 2006). In our study, the abundance of NET did not change from PRE to POST (Figure 2C–D; $P = 0.57$), suggesting stable catecholamine reuptake capacity across PRE and POST. Overall, these findings demonstrate that despite modest transcriptional changes, the catecholamine biosynthesis and handling machinery in SAT remain largely stable across the periparturient transition.

Parity-Dependent Remodeling of Catecholamine Handling in SAT

In contrast to limited period effects, pronounced parity-associated differences were observed in catecholamine-related proteins within SAT. With increasing parity, adipose tissue exhibited greater protein abundance of TH (Figure 2E; $P = 0.009$) and its activated form pTH_{Ser40} (Figure 2E; $P = 0.03$), indicating increased upstream catecholamine synthetic potential.

In contrast, DBH, the enzyme responsible for converting dopamine to norepinephrine, decreased with parity (Figure 2E; $P = 0.003$), suggesting a reduced capacity for local norepinephrine synthesis. Concurrently, the protein abundance of NET (Figure 2E; $P = 0.004$) increased with parity, consistent with enhanced catecholamine reuptake and tighter regulation of extracellular norepinephrine availability.

Although norepinephrine turnover was not assessed in this study, the coordinated changes in TH, pTH_{Ser40}, DBH,

and NET expression suggest parity-associated remodeling of catecholamine handling rather than increased norepinephrine synthesis in SAT of periparturient dairy cows. This pattern is consistent with altered sympathetic signaling dynamics in SAT of multiparous periparturient dairy cows (Eisenhofer et al., 2004).

Transition-Associated Changes in Adipokines and Lipogenic Gene Expression

Metabolic profiling revealed a marked negative energy balance and lipolysis during the transition from PRE to POST, increased circulating concentrations of NEFA and BHB, decreased insulin concentrations, and BCS loss (Table 3). Accordingly, from PRE to POST, there was a decrease in the gene expression of the adipokines leptin (*LEP*; Figure 3A; $P < 0.0004$) and adiponectin (*ADIPOQ*; Figure 3A; $P = 0.0002$), and the lipogenic markers, fatty acid synthase (*FASN*; Figure 3A; $P < 0.0001$) and peroxisome proliferator-activated receptor gamma (*PPARG*; Figure 3A; $P = 0.0009$) in SAT. Interestingly, the downstream enzyme in the lipogenesis pathway, diglycerol-acyltransferase 1 (*DGAT1*; Figure 3A; $P = 0.49$) did not change in expression between the 2 periods. This confirms that early lactation-associated negative energy balance increases lipolysis and decreases lipogenesis in adipose tissue (Kenéz et al., 2015).

Adrenergic Signaling and Posttranslational Regulation of Lipolysis

Catecholamine-mediated release of NEFA from adipose tissue is dependent on adrenergic receptors' abundance and posttranslational modification of lipolytic proteins, such as perilipin 1 (PLIN1) and HSL (Jaster and Wegner, 1981; Figure 3B). Among the adrenergic receptors, ADRB2 is considered the primary regulator of lipolysis in bovine adipose tissue (Ferland and Chilliard, 1999). In our study, ADRB2 protein abundance tended to decrease from PRE to POST (Figure 3C–D; $P = 0.06$), consistent with prior studies in early postpartum cows (Ferland and Chilliard, 1999, Rocco and McNamara, 2013). This decrease may reflect reduced adrenergic sensitivity during early lactation, despite elevated circulating catecholamines.

Downstream of adrenergic signaling, *PLIN1* gene expression decreased from PRE to POST (Figure 3A; $P = 0.03$), and PLIN1 protein abundance remained unchanged (Figure 3C–D; $P = 0.82$). Similarly, neither gene expression of hormone-sensitive lipase (*LIPE*; Figure 3A; $P = 0.55$) nor total HSL protein abundance (Figure 3C–D; $P = 0.62$) differed between periods. In contrast, phosphorylation of HSL at serine 660 tended to increase in POST (Figure 3C–D; $P = 0.11$), but no change in pro-

tein abundance of phosphorylated HSL at serine 563 was observed (Figure 3C–D; $P = 0.20$).

Parity exerted additional effects on HSL phosphorylation. Phosphorylation of HSL at serine 563 increased with parity (Figure 3E; $P = 0.03$), although this was not accompanied by a corresponding increase pHSL_{Ser563}:HSL ratio ($P = 0.35$; Supplemental Table S1, see Notes). The ratio of pHSL_{Ser660}:HSL remained unchanged by period (Figure 3C–D; $P = 0.33$) and parity ($P = 0.40$; Supplemental Table S1). These findings align with those of previous studies in dairy cows showing an increase of pHSL_{Ser563}:HSL from prepartum to the fresh period (De Koster et al., 2018). In agreement with our results, previous analysis of dairy cow tailhead SAT samples collected at 42 d prepartum and 1 d postpartum revealed an increased ratio of pHSL_{Ser563}:HSL (Kenéz et al., 2019). An additional study revealed no change in pHSL_{Ser660} of tailhead SAT between 21 d prepartum and 1 d postpartum, but in contrast to our results, showed a decrease in total HSL (Locher et al., 2011), suggesting a temporal hierarchy in HSL phosphorylation. Although longitudinal studies of pHSL_{Ser563} and pHSL_{Ser660} dynamics across lactation are lacking in dairy cows, our data suggest that differential phosphorylation of HSL may follow a hierarchical or sequential pattern during the periparturient period and with parity, as previously demonstrated in humans (Watt et al., 2006). Collectively, these data indicate that lipolytic regulation during the transition period is predominantly governed by posttranslational mechanisms and that parity may influence adipose tissue sensitivity to adrenergic stimulation.

Adipose Plasticity and Beiging-Related Marker Expression During the Transition Period

Dairy cows are notorious for having low circulating glucose and insulin concentrations during the periparturient period, a physiological hallmark of negative energy balance. This period is also associated with insulin resistance, which, if prolonged, can lead to excessive lipolysis and the onset of metabolic diseases (Zachut et al., 2013). An emerging area of adipose tissue biology in rodents and humans is the study of beige adipocytes, a thermogenically competent cell type residing within white adipose tissue depots. Although beige adipocytes are developmentally related to white adipocytes, they exhibit molecular and functional features more closely aligned with brown adipocytes (Harms and Seale, 2013).

In rodent models, stimulation of adrenergic receptor β_3 (ADRB3) promotes the conversion of white adipocytes toward a brown-like adipocyte phenotype through a process termed “beiging” (Granneman et al., 2005). This transition is characterized by reduced lipid storage, increased mitochondrial biogenesis, and enhanced oxida-

tive phosphorylation mediated by uncoupling protein 1 (UCP1, Figure 4A; Granneman et al., 2005; Harms and Seale, 2013). To determine whether similar mechanisms operate in dairy cows, we evaluated the expression of selected beiging-related markers in tailhead SAT during the periparturient period.

The protein abundance of ADRB3 in SAT decreased from PRE to POST (Figure 4B–C; $P = 0.03$), suggesting a reduced capacity for adrenergic-driven beiging signaling after calving. In murine models, ADRB3 activation stimulates PPARG coactivator 1 α (PGC-1 α), which cooperates with PPARG, EBF2, and PRDM16 to promote UCP1 expression and thermogenic programming (Nicholls and Locke, 1984; Seale et al., 2007; Park et al., 2015; Kuryłowicz and Puzianowska-Kuźnicka, 2020). In our dataset, *PGC1A* gene expression increased (Figure 4D; $P = 0.04$) and *PPARG* gene expression decreased markedly from PRE to POST (Figure 3A; $P < 0.0001$). Although these changes may appear contradictory, both factors participate in metabolic pathways beyond thermogenesis. Consistent with prior reports, *PPARG* downregulation likely reflects suppression of lipogenesis, whereas increased PGC1 α expression may support mitochondrial activity independent of classical thermogenic activation (Ito et al., 2024).

To further evaluate whether canonical beiging pathways were engaged, we quantified expression of additional beige-associated markers, including EBF2, a key transcriptional regulator of brown and beige adipocyte identity (Wang et al., 2014), and TMEM26, a beige-specific surface marker (Harms and Seale, 2013). Both markers were detectable in all PRE and POST SAT samples, indicating that tailhead SAT adipocytes retain the molecular capacity for beiging in dairy cows. However, the expression of both *EBF2* (Figure 4D; $P = 0.0011$) and *TMEM26* (Figure 4D; $P = 0.005$) decreased in POST versus PRE, suggesting a suppression of beige-associated transcriptional pathways during the lactation onset in dairy cows. These findings are consistent with rodent studies demonstrating depot-specific regulation of beiging during the periparturient period, with some depots favoring thermogenic activation and others prioritizing lipid mobilization (Wei et al., 2024).

At the protein level, EBF2 abundance did not differ between PRE and POST (Figure 4B–C; $P = 0.51$), and TMEM26 protein showed a tendency to decrease following calving (Figure 4B–C; $P = 0.09$). Similarly, protein abundance of PR domain-containing 16 (PRDM16), a master regulator of beige adipocyte differentiation (Seale et al., 2007), remained unchanged across periods (Figure 4B–C; $P = 0.41$). Parity did not broadly influence gene or protein expression of beiging markers; however, EBF2 gene expression decreased with parity (Supplemental Table S1, $P = 0.03$), and pro-

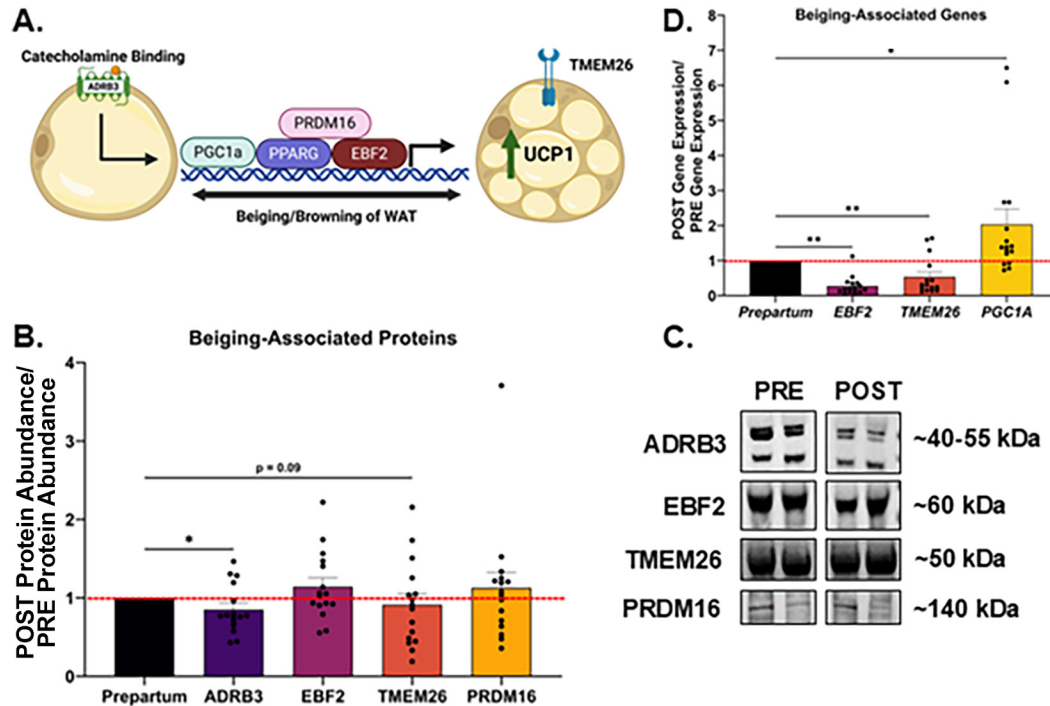


Figure 4. Effects on selected markers of adipose tissue beiging by parturition in the SAT of the PRE period (8.56 ± 2.3 d to parturition, mean $\pm$ SD, $n = 16$) and the POST period ($+7$ d after parturition, $n = 16$) of multiparous dairy cows (3.5 ± 1.6 lactations, $n = 16$). (A) Schematic diagram of beiging of white adipose tissue (WAT) known in mice. (B) Beiging-associated protein abundance of adrenergic receptor $\beta 3$ (ADRB3), early B-cell factor 2 (EBF2), transmembrane protein 26 (TMEM26), and PR domain-containing 16 (PRDM16). (C) Representative western blots of ADRB3, EBF2, TMEM26, and PRDM16. (D) Beiging-associated gene expression of *EBF2*, *TMEM26*, and PPARG coactivator 1 (*PGC1A*). Protein abundance and gene expression were analyzed by using a repeated-measures mixed-effects model with period by cow nesting within parity, and parity, period, and the interaction of parity and period as fixed effects and presented as a fold-change of postpartum measurement and prepartum measurement. * $P \leq 0.05$, ** $P < 0.01$. Biological schematic created in BioRender. Strieder-Barboza, C. (2026).

tein abundance tended to increase (Supplemental Table S1, $P = 0.09$), suggesting potential post-transcriptional regulation across successive lactations.

Ultimately, our dataset of the *UCP1* gene expression was undetectable in more than 75% of SAT samples, indicating that functional beige activation was largely absent in tailhead SAT, regardless of physiological stage. Collectively, these changes suggest that although SAT of periparturient cows retains phenotypic plasticity, classical beiging is not a dominant adaptive response during early lactation. The coordinated down-regulation of select beige-associated markers, coupled with unchanged PRDM16 abundance and the absence of *UCP1* expression, indicates that, if beiging occurs in this depot, it likely does not occur within the first 10 d following calving.

CONCLUSIONS

Our study advances understanding of adipose tissue biology in periparturient dairy cows by demonstrating that neural and catecholamine remodeling of tailhead SAT is more influenced by parity than by the acute periparturi-

ent transition from gestation to lactation itself. Despite marked metabolic changes from prepartum to postpartum, key components of catecholamine biosynthesis and handling remained largely stable at the protein level. In contrast, advancing parity was associated with increased abundance of neural and catecholaminergic markers, including TH, pTHSer40, PGP9.5/UCHL1, NET, NGFR, and GFAP, suggesting progressive expansion or reorganization of adipose tissue innervation with maturity. Downstream transcriptional changes in lipogenic and beiging-related pathways were primarily period-dependent and not indicative of active beige adipocyte programming. Collectively, these findings indicate that neural capacity within SAT of periparturient dairy cows is gradually remodeled across successive lactations and may contribute to metabolic adaptation independent of acute periparturient stress.

NOTES

This project was supported by the Agriculture and Food Research Initiative Competitive grant no. 2022-67015-36319 from the USDA-NIFA (U.S. Department of

Agriculture National Institute of Food and Agriculture, Washington, DC), 2021. Any mention of specific products in this article does not imply recommendation or endorsement of products by USDA. All opinions mentioned in this publication are solely those of the authors and do not reflect those of the USDA. The USDA is an equal opportunity lender, provider, and employer. The authors thank Hope Hall and her staff at the Texas Tech University Gordon W. Davis Meat Laboratory (Lubbock, TX) for their assistance in collecting beef nervous system samples. Additionally, the authors acknowledge Erminio Trevisi at the Università Cattolica del Sacro Cuore (Piacenza, Italy) for processing blood metabolites not analyzed by commercial kits. Supplemental material for this article is available at <https://doi.org/10.17632/wb366fft94.1>. The Institutional Animal Care and Use Committee at Virginia Tech approved all procedures for this study (protocol no. 22–071). The authors have not stated any conflicts of interest.

Nonstandard abbreviations used: AIC = Akaike information criterion; BIC = Bayesian information criterion; EL = elevated lipolysis; NE = norepinephrine; NEFA = nonesterified fatty acids; NGFR_{25kDa} = 25 kDa band of NGFR; NGFR_{55kDa} = 55 kDa band of NGFR; OL = optimal; POST = postpartum period; PRE = prepartum period; RIN = RNA integrity numbers; RT-qPCR = quantitative real-time PCR; SAT = subcutaneous adipose tissue; SNS = sympathetic nervous system; TBS = Tris-buffered saline; WAT = white adipose tissue.

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